

SATVIK KAUL

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PROFESSIONAL SUMMARY

AI and Machine Learning Engineer building production Generative AI, intelligent document processing, and agentic automation systems. Delivered multimodal extraction, RAG and AI-agent workflows, classification models, and secure ML applications using Python, LangGraph, FastAPI, MLflow, and cloud platforms.

PROFESSIONAL EXPERIENCE

Ampere Digital

November 2024 - Present

Machine Learning and AI Engineer

Remote

- Built a production Intelligent Document Processing pipeline using Claude multimodal extraction and asynchronous Cloudflare Queues, achieving 88% first-pass structured-field accuracy across 250+ aviation documents and reducing review time from ~7 minutes to under 60 seconds.
- Architected and deployed multi-model AI-agent workflows using RAG, LangChain, and LangGraph for onboarding, document review, and operational automation, reducing agent-turn latency by ~40% while supporting 1,200+ guided interactions per month.
- Trained a decision-support classifier on ~3,000 labeled review decisions; evaluated XGBoost, LightGBM, and CatBoost through MLflow experiments and applied a 0.97-precision threshold to automate ~60% of extracted-field decisions while routing uncertain cases for human review.
- Built AI evaluation and monitoring workflows using LLM-as-judge, OpenTelemetry, and Sentry to validate output quality, surface failures, and support reliable deployment of Generative AI applications.

Toronto Metropolitan University

September 2025 - Present

Research Assistant, Advanced Artificial Intelligence Lab (AI²)

Toronto, ON, Canada

- Built a secure end-to-end AI application using FastAPI and React for pathology image upload, algorithm selection, analysis-job submission, and report generation, with per-user isolation for uploaded data and results.
- Designed a modular architecture for integrating additional AI/ML algorithms and workflows without redesigning the core application, supporting future expansion of automated analysis capabilities.
- Evaluated AWS, Azure, and Google Cloud across 15 infrastructure configurations for a HIPAA/PHIPA-aligned deployment, recommending AWS based on cost, regional availability, and compliance maturity.
- Conducted self-supervised learning research using CAE, SimCLR, and MoCo v2 for four-class Gleason grading on SICAPv2, achieving 62.8% accuracy and 0.500 Cohen's kappa versus a 47.6% / 0.341 supervised baseline; work shortlisted for a conference, with manuscript in preparation.

Ministry of the Attorney General

January 2024 - October 2024

IT Analyst

Toronto, ON

- Led a three-person Agile team to translate legal and administrative stakeholder requirements into a Power Apps workflow-automation solution, improving process efficiency by 25% and reducing helpdesk escalations by 30%.

SKILLS

Programming: Python, SQL, Java, TypeScript, C/C++

Generative AI & Agents: Generative AI, AI Agents, LangChain, LangGraph, RAG, Anthropic Claude API, OpenAI API

Intelligent Document Processing: Document Understanding, Multimodal Extraction, Information Extraction, Structured Extraction, Information Retrieval, NLP

ML Engineering & MLOps: MLflow, Airflow, Docker, Automated Testing, GitHub Actions, OpenTelemetry, Sentry

Cloud & Services: AWS, Microsoft Azure, Cloudflare Workers, Cloudflare Queues, REST APIs, Asynchronous Processing

EDUCATION

Toronto Metropolitan University

September 2025 - August 2026

Master's, Computer Science

GPA: 4.28/4.33

Centennial College

September 2019 - June 2023

Bachelor's, Information Technology

GPA: 3.8/4.0